



▼ Programming

Programming means giving instructions to a computer to perform a task.

Programming Language

A programming language is a formal language used to communicate instructions to a computer
python, java, C, C++, JavaScript, C#, php, .net, ruby, Rust, Kotlin R. SQL

A- Low Level Language

- These are closer to machine hardware.
- Assembly language
- 010101011110101001010101

B - High Level Language

- easier for humans to read and write
- `Select * from HR;`

Compiled vs Interpreted Language

c, c++, rust - compiled

`code -- compiler --- execute --- run`

Interpreted language executes the code through a runtime rather than a compiler. ex- Mysql.

Syntax -

Syntax mean the rule for writing valid code-- .

- if you dont follow the rule - syntax error

IDE - Integrated Development Environment

An IDE is a software - that provides tools for writing, running, debugging and managing the program

- WorkBench
- VS CODE - visual studio code
- Jupyter Notebook
- Spyder
- ONLINE - GOOGLE COLAB .PY - PYTHON FILE - IDE .pynb - PYTHON NOTEBOOK .ipynb - Internet Python Notebook .pbix - powerbi file .excel - excel .sql - mysql .csv - comma seperated values .tsv - tab seperate values

extention - .jpeg , .pdf , .png

Python -

Python is a High level General purpose programming language wepage, website, games, AI , APP ,

- Easy to read and write

Variable - it is a container

- variable name is anything
- not starting with numbers
- variable name not contain space
- rather than use _

`tushar = 10 tushar1 = 10 1tushar = 10 ---- wrong tushar atkare = 10 -- wrong`

`tushar_atakre`

`tushar-atkare --- wrong`

- Python is a case sensitive language

Sudhiksha SUDHIKSHA SuDhiksha sudhikSHA

- The Sudhiksha is not defined

Data Types

int - 10 , 20 , 30 a = 10 float - 10.5 , 9.6 complex - 2 +3j string - str - "Tushar" boolien bool - True / False (true / fales is wrong)

list = [10,20,30,40,50,60] set = {10,20,30,50,60,} tuple = (10,20,30,50) dictionary - dict = {key:value}

NoneType- None

Comments - # single line comments

"""this is multi line comment """

Operators - symbols for calculations

- , - , * , /

// - floor division % - moduler operator (for reminder) ** - power

comparision Operator

, < , >= , <= , == (equal)

= - assign operator

a = 10

10 == 10

logical

NOT, AND, OR

keyword - A keyword is a reserve word in python with predefine meaning

True , False, If, Else, Elif, while, for, def, return, pass, continue, from, import, None, and , or, not , is , in

Function - A reusable block of code

SUM() , MIN() PRINT()

- predefine function - build in function
- user define function - (sangeet)
- function syntax is --- function_name() = MIN()

method - it is a funcion associated with an object

name = "tushar"

name.upper() - method -- syntacx - object.functionname()

Module -

- A module is a python file containing code such as function, methods, variable, classed , objects that can be import in another program
- every module need to import first - before using

import math

math.sqrt(25)

Library - A collection of reusable code that help prograaming to perform tasks

import matplotlib import seaborn import pandas import numpy import sklearn

before importing the lbrary we need to install it

pip - python installing package

python is an indented language

- indentation - Specific space use as a code in syntax

- indetation error

conditional statements

IF, ELSE, ELIF

LOOP - REPEAT CODE

- FOR -
- WHILE -

Debugging -

Bug - error debug - solve the error

Algorithm -

step - by step procedure for solving the problem



 Start coding or [generate](#) with AI.

[Colab paid products](#) - [Cancel contracts here](#)

 Variables  Terminal



 Python 3